

Engineering Physics (2025)

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Module 2 Unit 2: Optoelectronics

Course Instructor:

Dr. Sreekar Guddeti

Assistant Professor in Physics

Department of Science and Humanities

Vignan's Foundation for Science, Technology and Research

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p-n junction diode – forward and reverse conditions

Solar cell

Photo-diode

Direct and indirect band gap semiconductors

Light Emitting Diode

Tunneling diode

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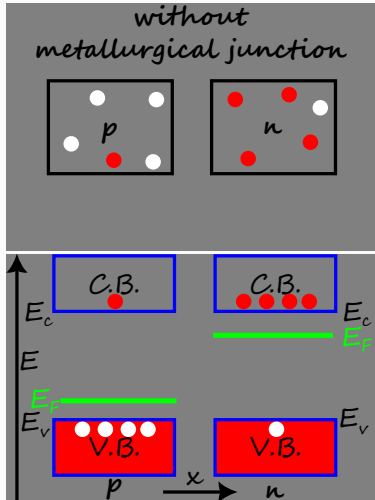
Direct and indirect band gap semiconductors

Light Emitting Diode

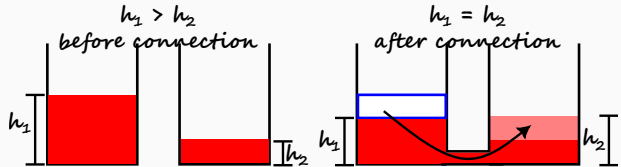
Tunneling diode

Isolated p-type and n-type semiconductors

- The Fermi level of p-type semiconductor is closer to the valence band edge E_v .
- The Fermi level of n-type semiconductor is closer to the conduction band edge E_c .



Analogy of gravitational force

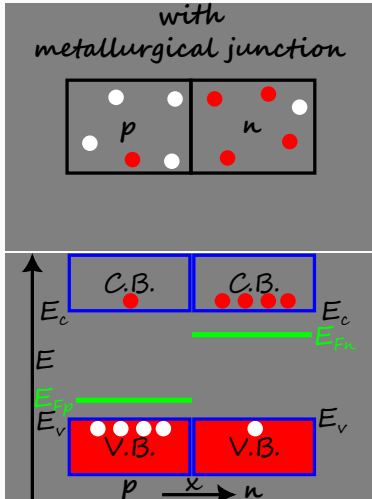


- When the glasses are connected, the the gravitational force acts as driving force to reach equilibrium.
- In the case of semiconductors, the driving force to reach equilibrium is E_F .

pn junction – Metallurgical junction

Definition

Metallurgical junction is defined as the interface between the p-type and n-type semiconductors.



- Upon the formation of metallurgical junction, there is a difference in the Fermi level in both the regions.
- The n region has higher Fermi level E_{Fn} than p region E_{Fp} , because there are more electrons in the n-region.

pn junction at zero bias

Definition

Zero bias condition is considered when external battery is not connected across p-region and n-region.

- The word voltage bias is used in the context of electricity to refer to external force.
- At zero voltage bias, there is **no application of external force** on the charge carriers.
- However, there are two internal forces at play –
 1. Diffusion forces due to gradients in concentration of charge carriers.
 2. Drift forces due to internal electric fields.
- The interplay of these internal forces leads to net zero force on charge carriers at thermal equilibrium.

Key Insight

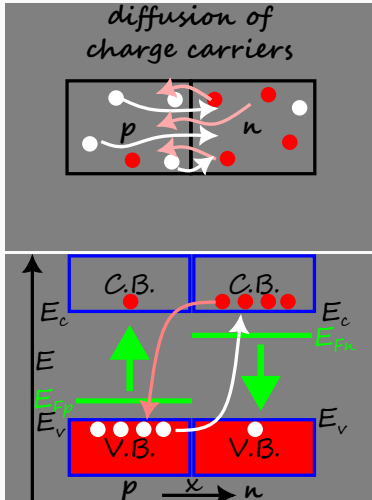
At zero bias, there is interplay of internal drift and diffusion forces.



pn junction – Diffusion

Definition

Diffusion is the net movement of particles from a region of higher concentration to a region of lower concentration.

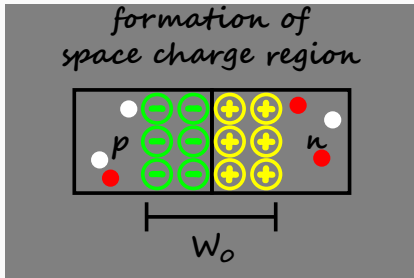


- The difference in Fermi level acts as a “driving force” so that electrons from n-region **diffuse** into the p-region.
- By similar analysis, holes from p-region diffuse into the n-region.
- Diffusion of electrons from n-region lowers its Fermi level E_{Fn} .
- Diffusion of holes from p-region raises its Fermi level E_{Fp} .

pn junction at zero bias – Space-charge region

Definition

- **Space charge region**, also called the **depletion region**, is the region created as a result of diffusion of free charge carriers across the metallurgical junction.
- The width of the depletion region is called depletion region width W_0 .



- Due to the diffusion of mobile charge carriers, their **ionic counterparts** are left without partners.
- Space charge region is also called the **depletion region** since the region is depleted ^a of the free charge carriers.

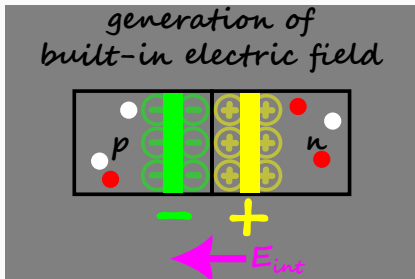
^adeplete means decrease or remove.

- The depletion region consists of positively charged donor ions on the n-side of the junction and negatively charged acceptor ions on the p-side of the junction.

pn junction at zero bias – Built-in electric field

Definition

Built-in electrical field, also called internal electric field E_{int} is the electric field generated due to the oppositely charged regions on either side of the metallurgical junction.

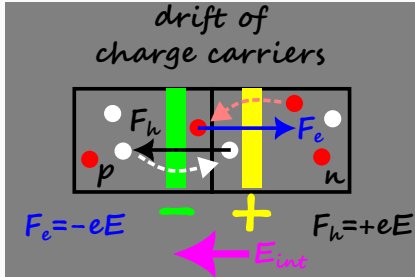


- Within the depletion region, the oppositely charged ions on either side of junction can be considered as plates of a capacitor.
 - Since the plates are charged, a **built-in electric field** is generated.
 - The built-in electric field is directed from **n-region to p-region**.
- E_{int} prevents further build up of space charge due to diffusion as it opposes the diffusion of electrons from n-region into p-region and vice versa.

pn junction at zero bias – Drift

Definition

Drift is the net movement of charged particles due to the application of electric field.



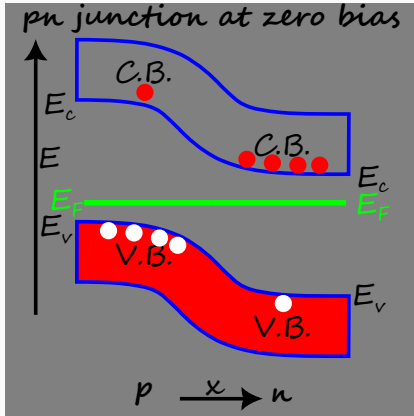
- Consider a diffused electron that is present in the p side of the depletion region.
- Since the built-in electric field is directed from n-region to p-region, the electric force is directed from p-region to n-region.
- Electron drift backs to the n-region.

- The diffusion force is counterbalanced by the drift force.
- Similar analysis applies to the diffused hole.

Problem

Prove the counterbalancing effect of drift and diffusion for holes.

pn junction at zero bias – Thermal equilibrium



Definition

Thermal equilibrium occurs when the net force on the system is zero.

- Since the difference in Fermi level is a “driving” force, when the system reaches thermal equilibrium, the net force is zero.
- This implies the Fermi level is a constant throughout the system at thermal equilibrium.

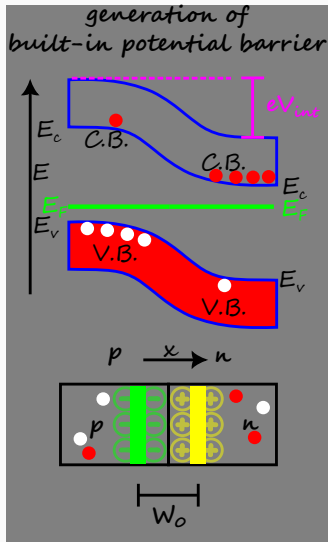
- Since the external force is zero at zero bias, the pn junction is at thermal equilibrium.

Key Insight

At thermal equilibrium, E_F is a constant.



pn junction at zero bias – Built-in potential barrier



Definition

Built-in potential barrier V_{int} is the potential difference generated across the oppositely charged halves of the depletion region.

- If we model the depletion region as a capacitor with electric field E_{int} between the plates and the separation between the plates is W , then the potential difference across the plates is given by

$$E_{int} = \frac{V_{int}}{W_0} \Rightarrow V_{int} = E_{int} W_0$$

Key Insight

Potential barrier is proportional to the built-in electric field.



pn junction at non-zero bias

pn junction – at non-zero bias

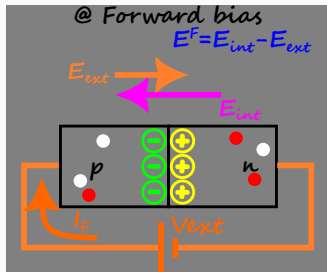
- At non-zero voltage bias, there is **an application of external force** on the charge carriers. As a result, the pn junction cannot attain thermal equilibrium.
- Since there is no thermal equilibrium, the Fermi energy is not a constant across the pn junction.
- Since the direction of built-in electric field is from n-region to p-region, the behaviour of p-n junction is different when the external field E_{ext} is parallel or anti-parallel to the built-in electric field E_{int} .
 1. $E_{\text{ext}} \uparrow \downarrow E_{\text{int}}$ corresponds to forward biasing.
 2. $E_{\text{ext}} \uparrow \uparrow E_{\text{int}}$ corresponds to reverse biasing.

Key Insight

At non zero bias, the Fermi energy is not a constant across the pn junction.



pn junction – at forward bias



Forward

$$V_F > 0$$

$$E^F < E_{\text{int}}$$

$$W_F < W_0$$

$$V_F < V_{\text{int}}$$

$$E_{F_n} - E_{F_p} > 0$$

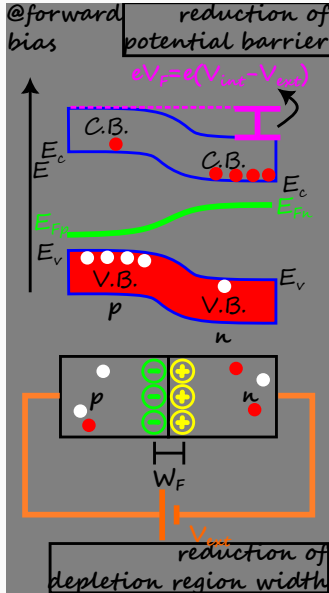
$$I_F > 0$$

Definition

Forward bias condition is considered when the positive electrode of battery with potential V_F is connected to p-region and negative electrode is connected to n-region.

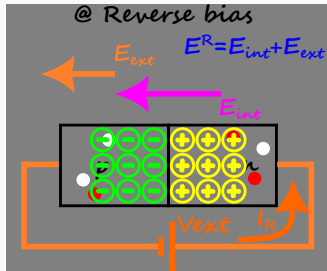
- The external bias $V_{\text{ext}} = V_F$ is from p-region to n-region.
- The external field E_{ext} is from p-region to n-region. The internal field E_{int} is **anti-parallel** to E_{ext} . The net electric field E^F is lower than the case of zero bias.
- So less space charge is created in depletion region. So we have smaller space charge width W_F .

pn junction – at forward bias



- Since potential is proportional to electric field, potential barrier V_F is lowered.
- From the band diagram, Fermi energy of n-region E_{Fn} is higher than Fermi energy of p-region E_{Fp} .
- So electrons will move from n region to p-region. This leads to current from p-region to n-region. This is called forward bias current I_F .
- Since electrons are present in n-region this leads to **good conduction** of pn-junction.

pn junction – at reverse bias



Reverse

$$V_R < 0$$

$$E^R > E_{\text{int}}$$

$$V_R > V_{\text{int}}$$

$$E_{Fn} - E_{Fp} < 0$$

$$W_R > W_0$$

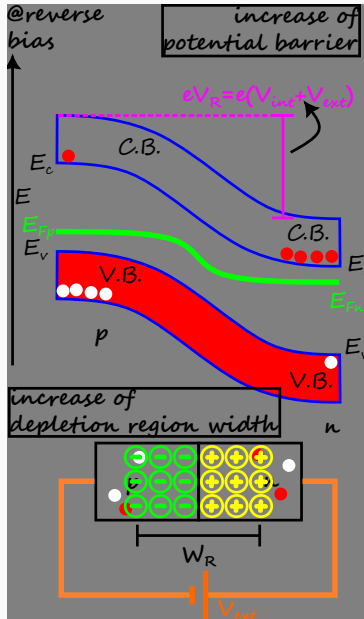
$$I_R < 0$$

Definition

Reverse bias condition is considered when the positive electrode of battery is connected to n-region and negative electrode is connected to p-region.

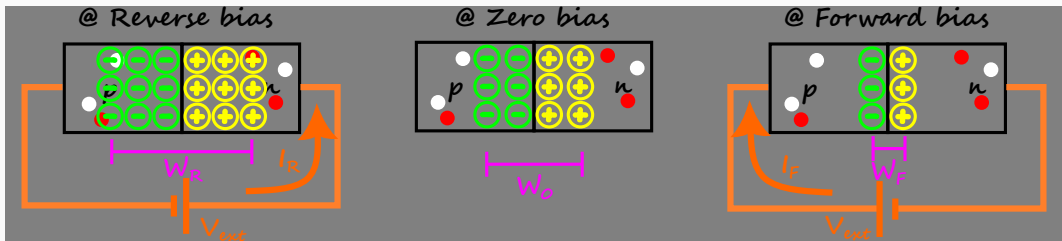
- The external bias $V_{\text{ext}} = V_R$ is from n-region to p-region.
- The external field is from n-region to p-region.
Under reverse bias condition, the internal field is **parallel** to the external field. The net electric field E^R is higher than the case of zero bias.
- So more space charge is created in depletion region. So we have larger space charge width W_R .

pn junction – at reverse bias



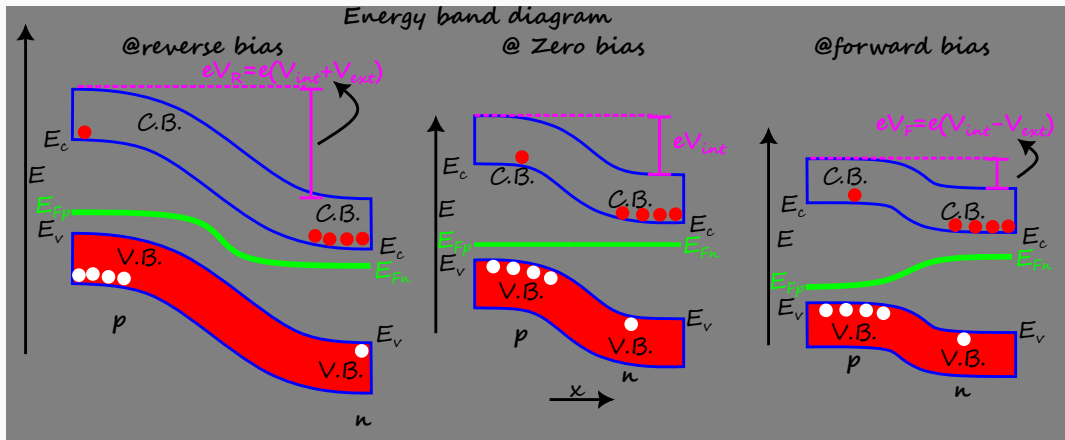
- Since potential is proportional to electric field, potential barrier V_R is raised.
- From the band diagram, Fermi energy of n-region E_{Fn} is lower than Fermi energy of p-region E_{Fp} .
- So electrons want to move from p region to n-region. But only minority carriers are there in p region.
- Since very few electrons are present in n-region this leads to **poor conduction** of pn-junction and the current is called reverse bias current I_R .

Comparison of depletion region widths at different biases



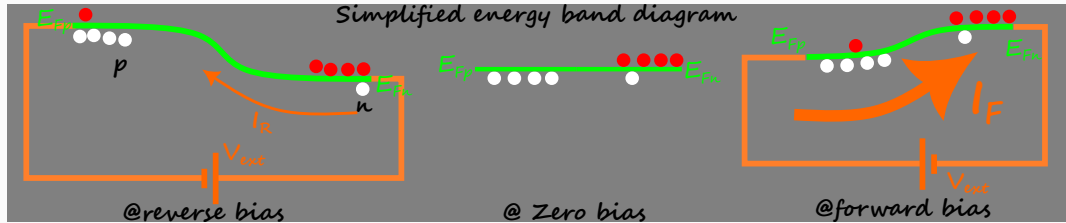
$$W_R > W_0 > W_F$$

Comparison of potential barrier at different biases



$$V_R > V_{int} > V_F$$

Concept of simplified energy band diagram

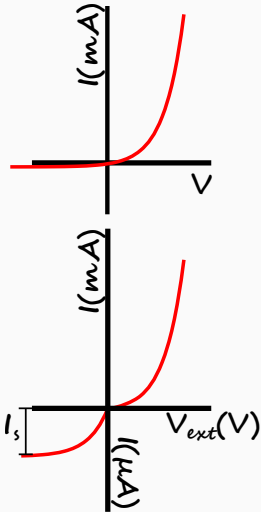


- Energy band diagram has many features. It is sometimes useful to represent the most essential things of the energy band diagram.
- We have used the potential energy barrier to understand motion of charge carriers. However, Fermi level can also be used for the analysis.
- The band diagram that contains only the variation of Fermi level along with charge carriers is called simplified band diagram.
- The electrons can be imagined to occupy the Fermi level on the top side and holes can be imagined to occupy the Fermi level on the bottom side.
- The **electrons want to reach lower levels** of Fermi energy whereas holes **want to reach higher levels** of Fermi energy.

Large forward current vs tiny reverse current

- Under biasing conditions, the electrodes of the battery change the Fermi levels of the p and n regions.
- Since the negative electrode supplies electrons, it raises the Fermi level of the connected region. Similarly, positive electrode lowers the Fermi level.
- Under forward biasing, the energy of electrons at n-region is raised. The flow of electrons to lower energy p-region is possible. Similarly, the energy of holes at p-region is lowered. The flow of holes to higher energy n-region is possible. This leads to **large forward current**.
- Under reverse biasing, the energy of electrons at n-region is lowered. The flow of electrons to higher energy p-region is difficult. Similarly, the energy of holes at p-region is raised. The flow of holes to lower energy p-region is difficult.
- However, the reverse bias is favourable for minority electrons in p region and minority holes in n-region. Since the number of these carriers is very few, we see a **tiny reverse current**.

pn junction – $I - V$ characteristics



Definition

Rectifying action of pn junction is its property to allow **uni-directional** conduction.

- The diode equation can be shown to be

$$I = I_s \left[\exp \left(\frac{eV}{k_B T} \right) - 1 \right]$$

where I_s is the reverse saturation current.

- The equation and the $I - V$ capture the non-linear nature of p-n junction.
- In the forward bias condition, it acts as a “closed switch” allowing large current to flow through it.
- In the reverse bias condition, it acts as an “open switch” allowing insignificantly low current to flow through it.
- This unidirectional conduction is called rectifying action.

Definition

The drift current due to minority carriers is known as **reverse saturation current**.

- As the reverse bias increases in magnitude, the reverse bias current saturates to a constant and is given by

$$\begin{aligned} & \lim_{V \rightarrow -\infty} I_s \left[\exp \left(\frac{eV}{k_B T} \right) - 1 \right] \\ &= I_s [\exp(-\infty) - 1] \\ &= I_s(0 - 1) \\ &= -I_s \end{aligned}$$

- This saturation current in the reverse bias direction is called reverse saturation current.
- Since the minority carrier concentration does not depend on the bias and only depends on the temperature, it is a constant at a given temperature.

Electrical power

Definition

Power is defined as the rate change of work done per unit time.

- The mechanical power P_{mech} is given by

$$\begin{aligned} P_{\text{mech}} &= \frac{dW}{dt} = \frac{dFs}{dt}, \\ &= F \frac{ds}{dt} = Fv. \end{aligned}$$

Here F is the force applied along displacement s to do work $W = Fs$ on a particle moving with velocity v .

- The electrical power consumed along a resistor P_{elec} for one electron is given by $P_{\text{elec}} = Fv = -eEv_d$ where v_d is the drift velocity.
- The power consumed within a length L and area A of resistor with electron concentration n is $P_{\text{elec}} = -nLAeEv_d = -VjA = -VI$ where $V = El$ is the potential drop across the resistor and $I = jA$ is current across the resistor. Here we have used definition of current density $j = nev_d$.

pn junction – Power characteristics

Definition

The sign convention employed for power consumption is if the current direction is along the voltage drop, then power is consumed and if the current direction is opposite the voltage drop, then power is generated.

	R	B	pn
V_F	> 0	> 0	> 0
I_F	> 0	< 0	$\gg 0$
P_F	> 0	< 0	$\gg 0$
V_R	< 0	N.A.	< 0
I_R	< 0	N.A.	< 0
P_R	> 0	N.A.	> 0

- For the resistor, the current is always along the voltage drop, even if the voltage drop is reversed. Therefore, the power is consumed in both biases.
- For the battery, the current is opposite to the voltage drop, Therefore, power is generated in a battery.
- For the pn junction, the current is along the voltage drop in both cases. Therefore, the power is consumed in both biases. However, the magnitudes are different unlike resistor.

Conceptual Questions

Problem

What causes majority carriers to flow at the moment when p-region and n-region are brought together? Why does this flow not continue until all carriers have recombined?

[Hint for second part: Convince yourself that the built-in electric field opposes the diffusion “forces” for both the electrons and holes.]

Tough problems

Problem

Prove that the analysis of pn junction using potential energy barrier maps with the analysis using Fermi energy level. In other words, prove that $E_{Fn} - E_{Fp} = eV_{\text{ext}}$.

Test Your Understanding

1. What causes majority carriers to flow at the moment when p-region and n-region are brought together? Why does this flow not continue until all carriers have recombined? [Hint for second part: Convince yourself that the built-in electric field opposes the diffusion “forces” for both the electrons and holes.]
2. Explain the formation of **potential barrier** across the pn junction region.
3. Explain the formation of **depletion region** in the pn junction region. What is the typical value of depletion region width for commonly used pn junctions?
4. Draw band diagrams for symmetrically doped pn junction at
 - zero bias
 - forward bias
 - reverse bias
5. What is reverse saturation current in a diode?
6. Why is reverse saturation current independent of reverse bias?
7. What is reverse breakdown?

Technical terms

- metallurgical junction
- diffusion
- built-in electric field
- thermal equilibrium
- built-in potential barrier
- depletion region
- space charge region
- space charge width
- reverse saturation current
- reverse breakdown

Solar cell

p-n junction diode – forward and reverse conditions

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Photo-diode

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Tunneling diode

Optoelectronics

- Electronics is the study of controlling electrical signals.
- Any electrical signal is characterized by its current and voltage properties.
- Electronics is broadly classified into three areas –
 - **Microelectronics** – here we study switching electrical signals, which is useful to represent binary logic. This is necessary for computation. This is also called **digital electronics**. Since power is not a concern, signals have currents of the order of microamperes (hence the name microelectronics) and voltages of the order of few volts
 - **Power electronics** – here we study amplifying electrical signals to transfer energy. This is necessary for energy requirements of society as it is well known that electrical energy is efficient way of energy transfer. Signals have currents of the order of hundreds of amperes and voltages of the order of thousands of volts. This is also called **analog electronics**.
 - Opto-electronics ¹ – here we harness the interaction between electron and photon. The advantages of light like its high velocity, free source of energy (Sun light), are combined with the advantages of electron like its easy manipulation to develop devices of specific requirements.

¹Opto-electronics is a combination of optics and electronics

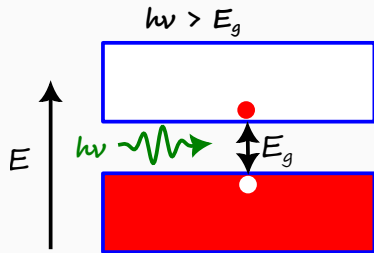
Definition

- A electrical device is a electrical component with a characteristic or signature $I - V$ behaviour.
- An opto-electronic device is an electrical device whose $I - V$ characteristic is determined by the interaction of light.
- The interaction of light with opto-electronic devices is of two-types –
 - optical absorption
 - Radiative recombination
- The opto-electronic devices based on optical absorption are
 - solar cell
 - photo-diode
- whereas the opto-electronic devices based on radiative recombination are
 - Light Emitting Diode (LED)
 - Laser Diode

Optical absorption

Definition

- Optical absorption is defined as the process of absorption of photon that leads to the generation of charge carriers.
- The generated electron-holes pairs are additional to the already present. Hence they are called **excess charge carriers**.



- In semiconductors, photon with energy greater than energy band gap E_g , it is absorbed and leads to generation of electron-hole pair.

Key Insight

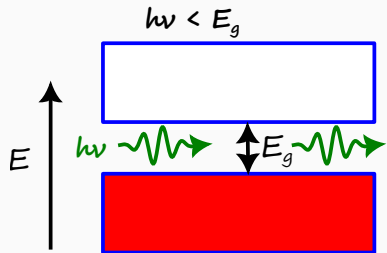
In optical absorption, the photon is the source of energy.



Optical transparency

Definition

Optical transparency is defined as the process of propagation of photon without the generation of charge carriers.



- If the photon energy is less than energy band gap, it is not absorbed and instead propagates through the semiconductor.

Problem

Why common glass is transparent to visible light.

[Hint: $E_g^{\text{SiO}_2} = 9 \text{ eV}$.]

Key Insight

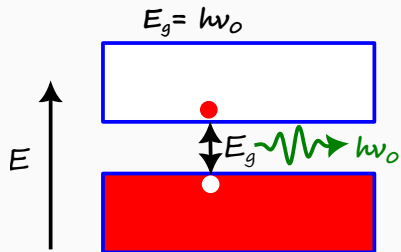
In optical transparency, the photon is not absorbed.



Radiative recombination

Definition

Radiative recombination is defined as the process of generation of photon upon combination of excess charge carriers.



- In semiconductors, electron-hole pair recombine to form the covalent bond and release the energy equal to energy band gap as a photon.

Key Insight

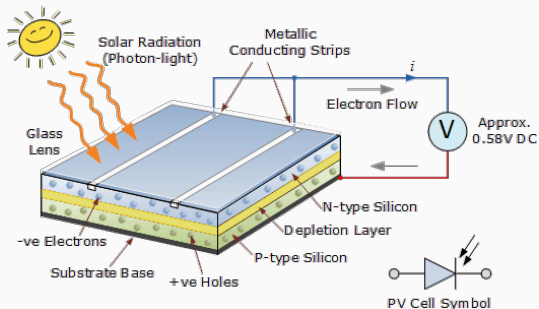
In radiative recombination, the charge carriers are the source of energy.



Solar cell – construction

Definition

- Solar cell, also technically called the **photovoltaic cell** is an opto-electronic device based on pn junction that can generate electrical power when illuminated.
- A collection of cells in series and parallel combination is called a solar panel



Sketch credit: Alternative Energy Tutorials

Sketch of solar panel. Sketch taken from Alternative Energy Tutorials

- The main aim of solar cell operation is to convert solar energy into electrical energy.
- Electrodes are deposited on both sides of junction. The top metal electrode is in the form of metal grid with fingers, which permits sunlight to pass through it.

Solar cell – working principle

Definition

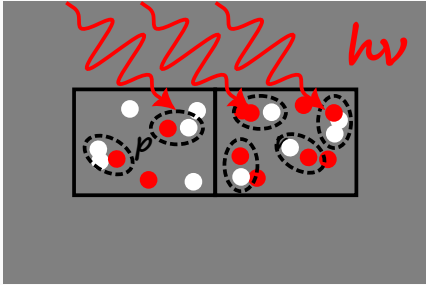
The working principle of solar cell involves three processes–

1. Generation of **excess** electron-hole pairs upon the illumination of light, and
2. Separation of electrons from holes **before** radiative recombination due to the built-in electric field.
3. Generation of voltage due to **accumulation** of excess separated charge carriers at the ends of the pn junction
 - The solar cell pn junction is operated without bias i.e. zero bias.
 - Upon illumination with photons of suitable energy, the photon are absorbed through the surface area. The covalent bonds of the lattice are broken releasing the free electrons and holes into the conduction band and valence band respectively.

Problem

A material has bandgap of 5 eV. Can it be used as solar cell material? Why or why not?

Working principle – 1. Excess electron-hole pairs



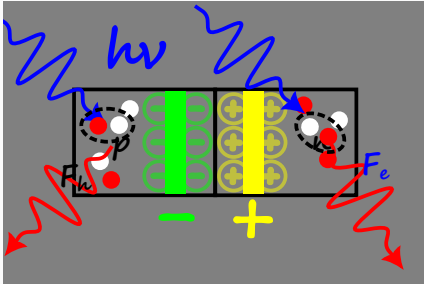
- Upon illumination with photons with energy greater than the band gap, optical absorption takes place.
- This results in the generation of excess electron-hole pairs throughout the device.

Key Insight

Optical absorption generates excess electron-hole pairs.



Working principle – 2a. Illumination away from depletion region



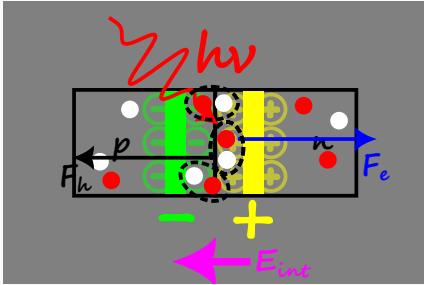
- In the neutral regions far away from the depletion region, the excess charge carriers again undergo radiative recombination and release photons.

Key Insight

Away from depletion region, generation of excess charge carriers is followed by radiative recombination.



Working principle – 2b. Illumination at depletion region



- At the depletion region also, the excess charge carriers are generated.
- However, before they undergo radiative recombination, the built-in electric field sweeps them out of the depletion.

Key Insight

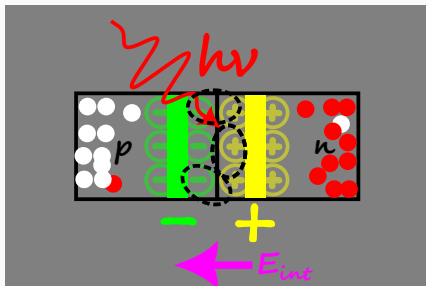
At the depletion region, separation of excess charge carriers prevents radiative recombination.



Working principle – 3. Photo-voltage

Definition

Photovoltage, also called photo e.m.f.² is the voltage appearing across the device due to the separation of excess charge carriers.



- The swept excess electrons and holes **accumulate** at the n region and p region edges respectively
- The accumulation of positive charge at the p-edge and negative charge at the n-edge leads to the generation of a voltage V_L .

- Since the source of voltage is illumination of photons, it is called **photovoltage**.

Key Insight

The photovoltage forward biases the pn junction.

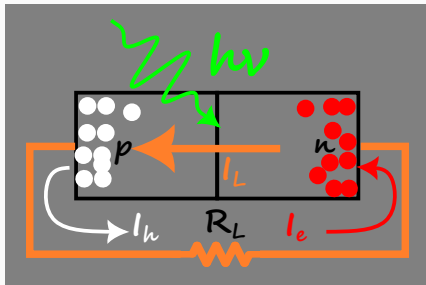


²e.m.f. stands for electro motive force

Solar cell – Photocurrent

Definition

Photocurrent is the current flowing through the solar cell when it is connected across a load resistance.



- Solar cell can be used to drive currents through loads.
- Upon connection to a load resistance R_L , the accumulated holes at the p-edge flow anti-clockwise whereas the accumulated electrons at the n-edge flow clockwise.

- Clockwise flow of holes leads to clockwise hole current I_h whereas anti-clockwise flow of electrons leads to clockwise electron current I_e .
- The net current I_L is called photocurrent ³ and is given by $I_L = I_h + I_e$.

Key Insight

The photocurrent flows from n to p region.

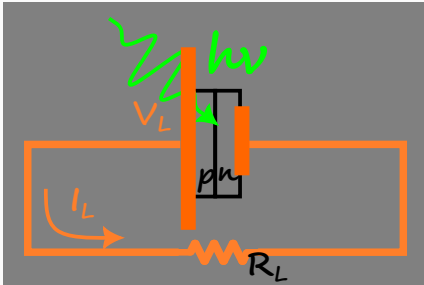
³suffix L stands for light and **not** load.



Solar cell – Power source

Definition

Photocurrent is the current flowing through the solar cell when it is connected across a load resistance.



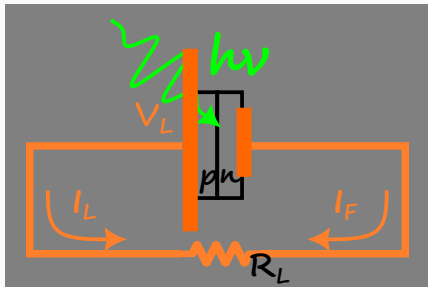
- The photo-voltage drop V_L is from p-n region.
- The photo-current I_L is from n-p region.
- Since the voltage drop is opposite the current direction across the solar cell, the power dissipated is negative.
- In other words, the **power generated is positive** and solar cell acts like a battery.

Key Insight

Solar cell under illumination acts as a power source.



Solar cell – Load biasing



- However, the photo-voltage also forward biases the p-n junction.
- This results in a forward bias current I_F in the anti-clockwise direction.
- The net current through the load resistance is

$$I_{\text{net}} = I_L - I_F$$

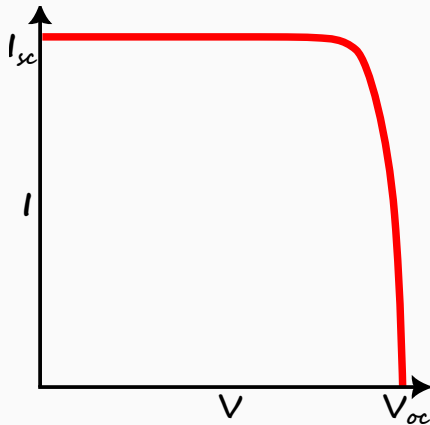
- Though the solar cell is operating under zero bias, upon connection to load, the load forward biases the pn junction.

Key Insight

Load forward biases the illuminated solar cell.



Solar cell – $I - V$ characteristics



- From Kirchoff voltage law, the photo-voltage is given by

$$V_L = I_L R_L$$

- From diode $I - V$ characteristics, the forward current I_F is given by

$$I_F = I_s \left[\exp \left(\frac{eV_L}{k_B T} \right) - 1 \right]$$

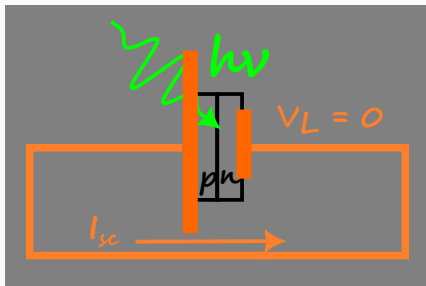
- Therefore the solar cell $I - V$ characteristics is given by

$$\begin{aligned} I_{\text{net}} &= I_L - I_F \\ &= I_L - I_s \left[\exp \left(\frac{eV_L}{k_B T} \right) - 1 \right] \end{aligned}$$

Solar cell – short circuit current

Definition

Short circuit current is defined the current through solar cell when it is short circuited or zero load resistance is connected.



- When the load resistance is zero, the solar cell generates maximum current. This condition is called short circuit condition.
- At short circuit condition, $R_L = 0$, so the voltage drop across load is zero

$$V_L = I_L R_L = 0$$

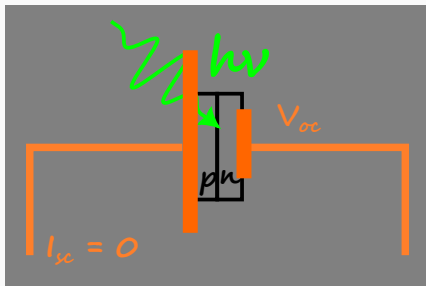
- The net current is the short circuit current I_{sc} and is given by

$$\begin{aligned} I_{sc} &= I_L - I_s (e^0 - 1) \\ &= I_L \end{aligned}$$

Solar cell – open circuit voltage

Definition

Open circuit voltage is defined the voltage across solar cell when it is open circuited or infinite load resistance is connected.



- When the load resistance is infinite, the solar cell generates maximum voltage. This condition is called open circuit condition.
- At open circuit condition, $R_L = \infty$, so the current is zero

$$I_L = \frac{V_L}{R} = 0$$

- The photovoltage is the open circuit voltage V_{oc} and is given by

$$I_L - I_s \left[\exp \left(\frac{eV_{oc}}{k_B T} \right) - 1 \right] = 0 \Rightarrow I_L = I_s \left[\exp \left(\frac{eV_{oc}}{k_B T} \right) - 1 \right]$$
$$\exp \left(\frac{eV_{oc}}{k_B T} \right) = \frac{I_L}{I_s} + 1 \Rightarrow V_{oc} = \left(\frac{k_B T}{e} \right) \ln \left(1 + \frac{I_L}{I_s} \right)$$

Solar cell – Power optimization

- A solar cell is a power generating device. However, at the limiting conditions, the power generated is zero.
- At short circuit condition, the power generated is

$$P_{sc} = V_{sc} I_{sc} = 0 \cdot I_{sc} = 0$$

- Similarly, at open circuit condition, the power generated is

$$P_{oc} = V_{oc} I_{oc} = V_{oc} \cdot 0 = 0$$

- The electrical power generated is a function of drop across load resistance $V = IR_L$ and is given by

$$P = V \cdot I = V \cdot \left[I_L - I_s \left\{ \exp \left(\frac{V_L}{V_t} \right) - 1 \right\} \right],$$

where

$$V_t = \frac{k_B T}{e}$$

is the thermal voltage associated with thermal energy.

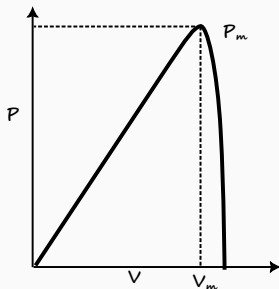
Solar cell – Power optimization

- To maximize P with respect to V , we need to find V_m , the voltage across the load when power is maximized. This is given by

$$\begin{aligned}\frac{dP}{dV} \Big|_{V_m} &= 0, \\ \Leftrightarrow I_L - I_s \left\{ \exp \left(\frac{V}{V_t} \right) - 1 \right\} - I_s \frac{V}{V_t} \exp \left(\frac{V}{V_t} \right) \Big|_{V_m} &= 0, \\ \Leftrightarrow I_L + I_s - I_s \exp \left(\frac{V_m}{V_t} \right) \left(1 + \frac{V_m}{V_t} \right) &= 0, \\ \Leftrightarrow \left(1 + \frac{V_m}{V_t} \right) \exp \left(\frac{V_m}{V_t} \right) &= \frac{I_L + I_s}{I_s}. \\ \therefore \left(1 + \frac{V_m}{V_t} \right) \exp \left(\frac{V_m}{V_t} \right) &= 1 + \frac{I_L}{I_s}\end{aligned}$$

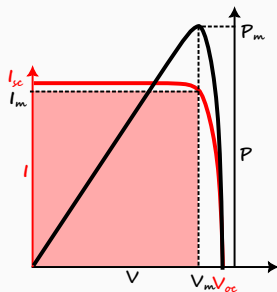
- The value of V_m can be determined by trial and error.

Solar cell – $P - V$ characteristics



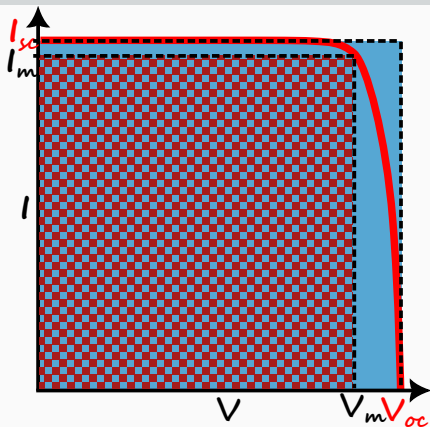
- The power generated in solar cell is completely consumed in the load resistance.
- If the photocurrent at V_m is I_m then the maximum power is

$$P_m = V_m I_m$$



- In $P - V$ curve, this is represented as (V_m, P_m) .
- The I & $P - V$ curves show the mappings of $I - V$ and $P - V$ curves. In $P - V$ curve, maximum power is represented as as red rectangle.

Output electrical power – Fill factor



- From the $I - V$ characteristics of solar cell, it is evident that the maximum current generated cannot exceed the short circuit current I_{sc} and the maximum voltage generated cannot exceed the open circuit voltage V_{oc} .
- The theoretically possible maximum power that can be generated should be less than the product of I_{sc} and V_{oc} . In the $I - V$ curve, this is represented as blue rectangle.

- The ratio of experimentally possible maximum power to theoretically possible maximum power is called Fill Factor and is given by

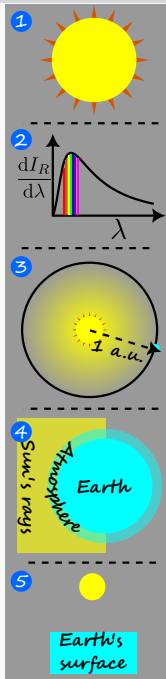
$$FF = \frac{V_m I_m}{V_{oc} I_{sc}}$$

Key Insight

Fill factor is a material property.



Input optical power – Solar irradiance



Definition

Solar irradiance is the solar power incident on the Earth's surface per unit area.

1. Sun as blackbody – Since sun is the major source of light energy, the input power is determined by the properties of sun.
2. Planck's blackbody radiation law – Sun is emitting photons of various wavelengths per unit time. Each photon with a given wavelength has an energy given by the Planck's law.
3. Sun -Earth separation – However, we are only receiving a fraction of the Sun's power.
4. Solar irradiance outside atmosphere – It is 1360 W m^{-2} .
5. Solar irradiance at Earth's surface – It is nearly 1000 W m^{-2} at noon due to losses in atmosphere.

Practical considerations – Solar cell efficiency

Definition

Conversion efficiency is defined as ratio of output electrical power to input optical power.

- To harness solar energy, the following practical limitations need to be considered.
- All the power cannot be harnessed as the optical absorption by a solar cell has a cut off wavelength due to the energy band gap.
- The photons with energy greater than energy band gap also generate electron-hole pairs. However, the difference of photon energy and band gap is dissipated as heat.
- The sum total of all these considerations is that a fraction of incident optical power is converted into electrical power and is given by the conversion efficiency η as

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100$$

Solar cell – applications

Chandrayaan 2 mission at glance [Click for video]



Image credits: ISRO

Problem

From the data in above figure, estimate the conversion efficiency of material used in Pragyan rover.

Problems on solar cell

Problem

A silicon solar cell under standard test conditions has $I_{sc} = 35 \text{ mA cm}^{-2}$, $V_{oc} = 0.60 \text{ V}$, Fill Factor (FF) = 0.80. The incident power is 100 mW cm^{-2} . Calculate the efficiency of the cell.

Problem

A solar panel has an efficiency of 18 % under standard irradiance of 1000 W m^{-2} . What panel area is required to generate 200 W of electrical power?

Problem

Each silicon solar cell has an open-circuit voltage of 0.60 V under illumination. How many such cells are needed in series to obtain an open-circuit voltage of approximately 18 V?

Problems on solar cell (Fun problems)

Problem

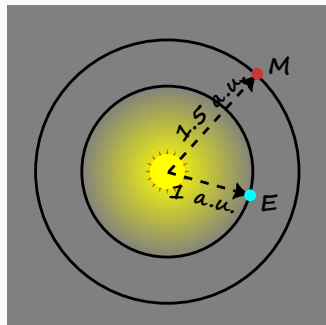
An alien ship crashlanded onto Earth's surface. From the debris, the solar panel used for power generation is found to have a band gap of 5 eV. Estimate the temperature of the alien Sun. [Hint: Solve below problem.]

Problem

Find the surface temperature of our Sun. For our sun, $\lambda_{\max}$ of blackbody body radiation spectrum is 510 nm (green), i.e., the greatest part of its radiation lies within the visible region of the spectrum. [Hint: Wien's displacement law ($\lambda_{\max} T = 2.898 \times 10^{-3} \text{ m K.}$)]

Problem

The power rating of solar panels of Pragyan rover is 50 W. If ISRO is successful to transport it onto Mars, what will be its Martian power rating?



Tough problems

Problem

The reverse saturation current density for a solar cell pn junction is

$J_s = 3.6 \times 10^{-11} \text{ A cm}^{-2}$. What is the photocurrent density required to produce open circuit voltage of 0.60 V?

Problem

For the same solar material used in above problem, the fill factor is 0.8. What is the maximum power per unit area generated if the photocurrent density is 150 mA cm^{-2} ?

Test Your Understanding

1. Explain the working of solar cell.
2. Draw and explain the $I - V$ characteristics of a solar cell.
3. A material has bandgap of 5 eV. Can it be used as solar cell material? Why or why not?
4. An alien ship crashlanded onto Earth's surface. From the debris, the solar panel used for power generation is found to have a band gap of 5 eV. Estimate the temperature of the alien Sun.

- optical absorption
- generation of excess charge carriers
- conversion efficiency
- fill factor
- open-circuit voltage
- short-circuit current

Photo-diode

p-n junction diode – forward and reverse conditions

Solar cell

Photo-diode

Direct and indirect band gap semiconductors

Light Emitting Diode

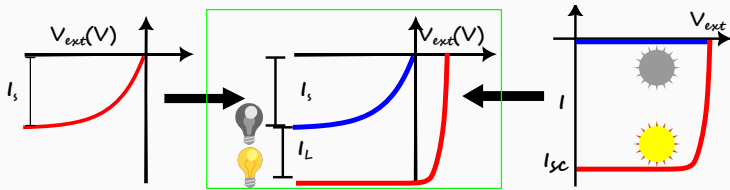
Tunneling diode

Definition

A photo-diode is an opto-electronic device that is used to sense the presence of light.

- A photo-diode is a pn junction diode that operates under **reverse bias**.
- Due to the reverse bias, the depletion region width increase. This leads to decrease in junction capacitance.

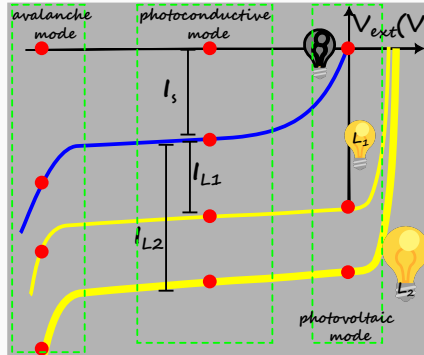
Photo-diode – $I - V$ characteristics



- The $I - V$ characteristics of photodiode can be considered as $I - V$ characteristics of solar cell under forward bias and $I - V$ characteristics of pn junction under reverse bias.
- The reverse bias generates reverse current that saturates to reverse saturation current I_s at high bias.
- In addition to the reverse current, if the photons are incident on the diode there is an additional photocurrent I_L .
- Thus the external bias is negative and the net current is negative and the $I - V$ characteristics is given by

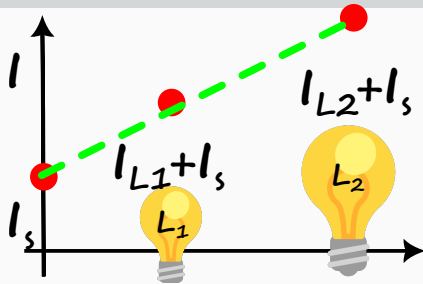
$$I = I_s \left[\exp \left(\frac{eV}{k_B T} \right) - 1 \right] - I_L$$

Photo-diode – Modes of operation



- The photo diode has three modes of operation
 1. photoconductive mode – reverse bias
 2. zero bias mode – reverse bias voltage is set to zero
 3. avalanche mode – **very** high reverse bias
- In every mode of operation, there are two states of operation
 1. Photons are not incident 💡
 2. Photons are incident 💡

Photoconductive mode



- Photo diode is under reverse bias.
- At zero light intensity, there is a non zero current due to the reverse saturation current. This is called **dark current**.
- Response of photodiode is linear. In this mode, photodiode is a linear sensor.

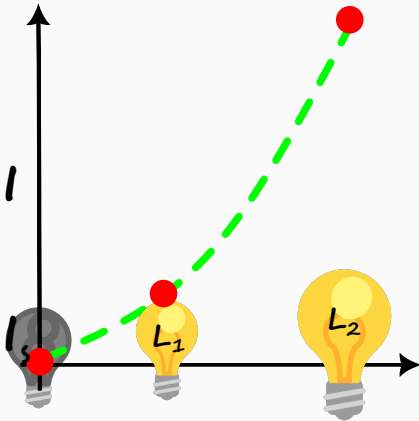
- Due to reverse bias, the depletion region width increases. Since the space charge region acts as a parallel plate capacitor, the junction capacitance C_{junction} is given by

$$C_{\text{junction}} = \frac{\epsilon A}{W}$$

where A is the cross sectional area of the depletion region and W is the depletion region width.

- Due to reduction in junction capacitance, photodiode is used for high frequency opto-electronic converters.

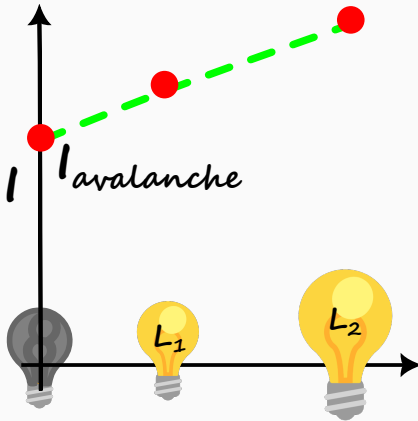
Photovoltaic mode



- Photo diode is under zero bias. This mode is the same as solar cell operation mode.
- At zero light intensity, there is zero current as there is no bias. Therefore, dark current is zero.
- Response of photodiode is non-linear. In this mode, photodiode is a non-linear sensor.

- Due to the zero bias, dark current is zero. So photo diode is used for detecting ultra low light.

Avalanche mode



- Photo diode is under very high reverse bias.
- The net electric field is so high that the generated charge carriers at the depletion region are accelerated to high kinetic energy. These energetic particles collide with silicon atoms and ionize them releasing secondary electrons. This is called **impact ionization**.
- The secondary electrons are also accelerated and ionize further atoms leading to a chain reaction also called avalanche.

- In this mode, photo diode is not used as sensor, but as an amplifier.
- Due to the avalanche effect, there is multiplication of the charge carriers. Therefore there is a net gain of current. In this mode, photo diode is used as an opto-electronic amplifier.

Direct and indirect band gap semiconductors

p-n junction diode – forward and reverse conditions

Solar cell

Photo-diode

Direct and indirect band gap semiconductors

Light Emitting Diode

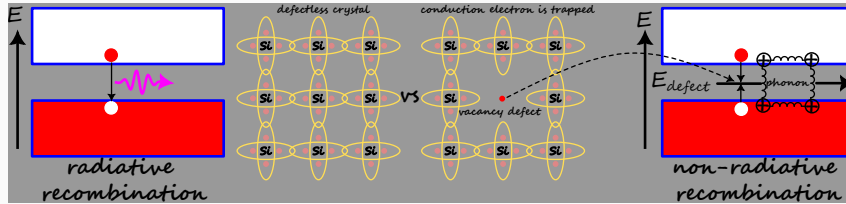
Tunneling diode

Definition

Luminescence refers to the general property of light emission.

- The previous two opto-electronic devices – solar cell and photo diode – were based on generation of excess carriers. The source of energy for generation was photon.
- The counterpart of generation is recombination.
- Recombination is of two types –
 - radiative recombination – the charge carriers recombine to emit photon
 - non radiative recombination – the charge carriers recombine to emit phonon i.e. heat.
- Here there are two types of radiative recombination –
 - **Photo luminescence** – If the excess charge carriers were generated by photon absorption, the photo emission from the recombination is called photo luminescence
 - **Electro-luminescence** – If the excess charge carriers were generated by current caused by an applied electric field, the photo emission from the recombination is called electro-luminescence.

Radiative vs Non radiative recombination



- Defects are imperfections in semiconductor. These are different from impurities.
- Some examples of defects are absence of atom in lattice – vacancy defect or presence of extra atom – interstitial defect
- Defects arise when temperature is increased. These lead to defect levels **within** the forbidden band gap.
- If the crystal is without defects, radiative recombination is preferred and final particle emitted is a **photon**.
- If the crystal is with defects, electron and hole reach the defect level first, then they recombine non radiatively and final particle emitted is a **phonon** or lattice vibration.

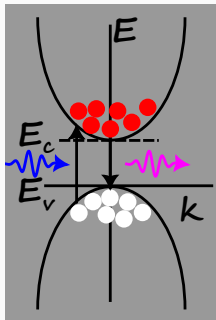
Radiative vs Non radiative recombination (contd...)

	Radiative recombination	Non-radiative recombination
Final particle	Photon is emitted	Phonon is emitted
Defectless semiconductor	Easy	Difficult
Defect states	Difficult	Easy

Direct band gap semiconductor

Definition

A direct band gap semiconductor is defined as a semiconductor with the top of valence band and bottom of conduction band at the **same** value of wave vector.



- Consider electron moves from E_c to E_v .
- In the radiative recombination process,
 - Energy of electron is transferred to energy of photon.
 - Initial momentum of electron is equal to final momentum.
 - Therefore, both the energy and momentum of the (electron + hole + photon) system is **conserved**.
 - **three** particles are involved



Key Insight

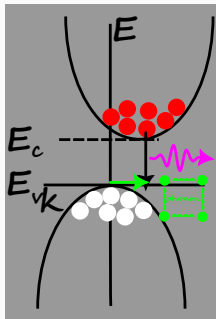
Radiative type of recombination is the dominant mechanism in a direct bandgap semiconductor.



Indirect band gap semiconductor

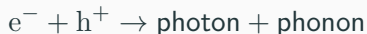
Definition

An indirect band gap semiconductor is defined as a semiconductor with the top of valence band and bottom of conduction band at the **different** values of wave vector.

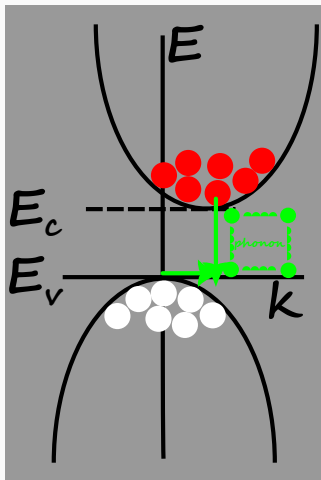


- Consider electron moves from E_c to E_v .
- In the radiative recombination process,
 - Energy of electron is transferred to energy of photon.
 - Initial momentum of electron is not equal to final momentum.
 - Therefore, energy of the (electron + hole + photon) system is **conserved** but the momentum is not conserved.

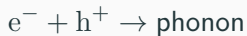
- To conserve momentum, a **phonon assisted radiative recombination** occurs where difference in momentum is supplied by phonon and **four** particles are involved



Indirect band gap semiconductor



- Since the four particle process is difficult to occur, the phonon-assisted radiative recombination is not dominant mechanism.
- Instead the photon emission is avoided and the energy and momentum conservation is satisfied by emission of phonon only.



- Emission of phonon leads to rise in temperature of material.

Key Insight

Non-radiative type of recombination is the dominant mechanism in an indirect bandgap semiconductor.



Direct vs indirect bandgap semiconductor

SNo.	Property	Direct	Indirect
1	k at E_C vs k at E_V	equal	unequal
2	Recombination mechanism	Radiative	Non-radiative
3	Particle emitted	Photon	Phonon
4	Periodic table	Compound s.c. like GaAs, InP	Elemental s.c. like Si, Ge
5	Applications	LED, laser diode	transistors, diodes

Light Emitting Diode

p-n junction diode – forward and reverse conditions

Solar cell

Photo-diode

Direct and indirect band gap semiconductors

Light Emitting Diode

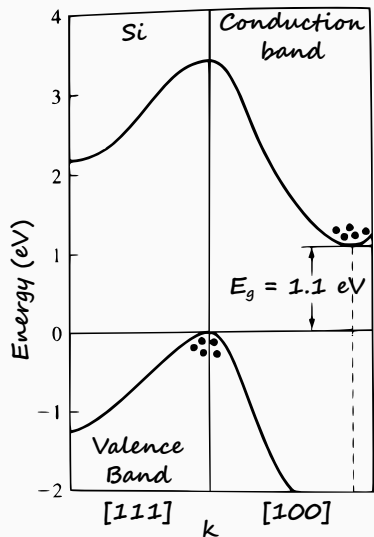
Tunneling diode

Definition

Light Emitting Diode (LED) is the opto-electronic device based on radiative recombination of charge carriers.

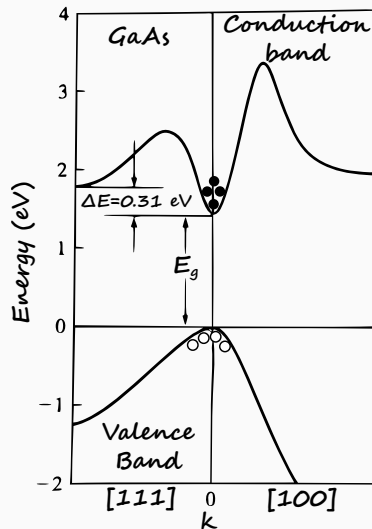
- LED is based on pn junction diode under forward bias.
- Under forward bias, the electrons and holes crossing the depletion region undergo radiative recombination leading to generation of light.
- From the perspective of electroluminescence, it is useful to use a semiconductor with a direct band gap semiconductor.

Elemental semiconductors – Si



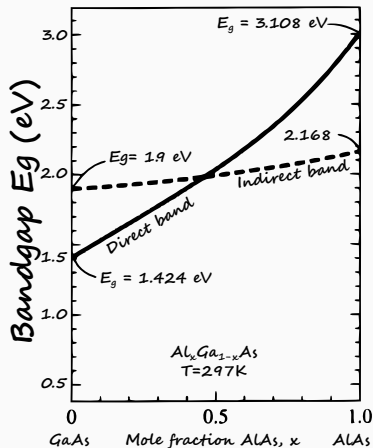
- Si is an elemental semiconductor.
- It is an indirect band gap semiconductor.
- The valence band maximum and the conduction band minimum occur at different values of wavevector.

Binary semiconductors – GaAs



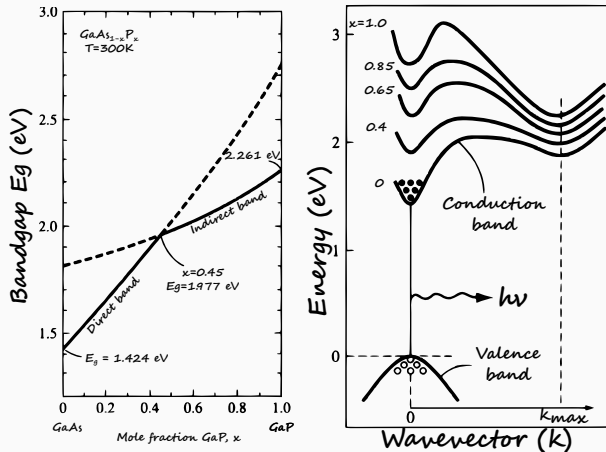
- GaAs is a binary semiconductor.
- It is a direct band gap semiconductor.
- The valence band maximum and the conduction band minimum occur at same value of wavevector $k = 0$.

Ternary semiconductors – $\text{Al}_x\text{Ga}_{1-x}\text{As}$



- $\text{Al}_x\text{Ga}_{1-x}\text{As}$ is a ternary semiconductor.
- It can be considered as a binary alloy of GaAs and AlAs. If the mole fraction of AlAs is considered as x and mole fraction of GaAs as $1 - x$, then the chemical formula is $\text{Al}_x\text{Ga}_{1-x}\text{As}$. Therefore $x = 0$ corresponds to pure GaAs whereas $x = 1$ corresponds to pure AlAs.
- From $x = 0$ to $x = 0.45$, the alloy is a direct band gap material. From $x = 0.45$ to $x = 1$, the alloy is an indirect band gap material.
- In the energy band gap vs mole fraction, there is a cross over from direct band gap to indirect band gap at $x = 0.45$.

Ternary semiconductors – $\text{GaAs}_{1-x}\text{P}_x$

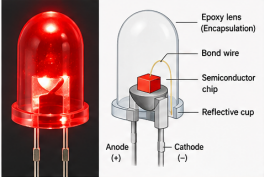
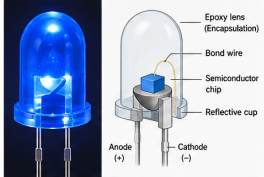
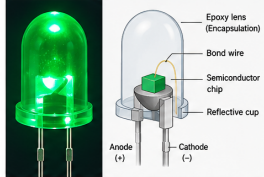
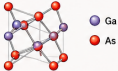
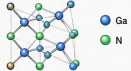
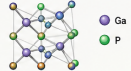


- $\text{GaAs}_{1-x}\text{P}_x$ is a ternary semiconductor with mole fraction x of GaAs and mole fraction $1 - x$ of GaP.
- From $x = 0$ to $x = 0.45$, the alloy is a direct band gap material. From $x = 0.45$ to $x = 1$, the alloy is an indirect band gap material.

- In the bandstructure, the conduction band shifts from $k = 0$ to $k = k_{\text{max}}$ as the mole fraction of GaP crosses $x = 0.45$.

LED – construction

LED COLORS AND SEMICONDUCTOR MATERIALS USED

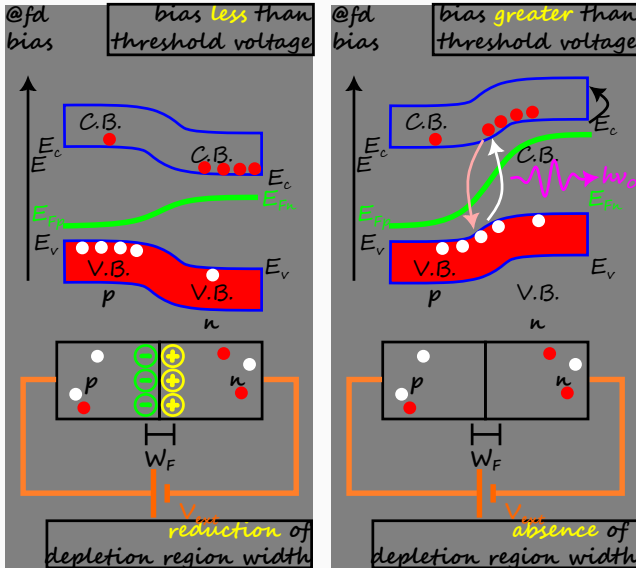
RED LED	BLUE LED	GREEN LED
		
MATERIAL USED GaAs (Gallium Arsenide) 	MATERIAL USED GaN (Gallium Nitride) 	MATERIAL USED GaP (Gallium Phosphide) (sometimes with Zn, O for better efficiency) 
EMITTED COLOR : Red WAVELENGTH : ~620 – 750 nm BAND GAP : ~1.8 – 2.0 eV	EMITTED COLOR : Blue WAVELENGTH : ~450 – 495 nm BAND GAP : ~2.7 – 3.4 eV	EMITTED COLOR : Green WAVELENGTH : ~495 – 570 nm BAND GAP : ~2.1 – 2.3 eV
NOTES GaAs is a direct band gap semiconductor that emits red light when electrons and holes recombine.	NOTES GaN is a wide band gap semiconductor that emits blue light.	NOTES GaP (and its alloys) are used to produce green light in LEDs.



SUMMARY | LED color depends on the semiconductor material and its band gap.
Higher band gap → shorter wavelength (blue) | Lower band gap → longer wavelength (red)

- New materials AlGaInP for Red, $\text{In}_x\text{Ga}_{1-x}\text{N}$ for Green, and $\text{In}_x\text{Ga}_{1-x}\text{N}$ for Blue.

LED – working principle



- LED works on the principle of **electroluminescence**.
- Below threshold voltage, potential energy barrier is reduced, and depletion region width is reduced.
- Above threshold voltage, potential energy barrier is removed, and depletion region width is removed. The electrons and holes move towards the junction, and **recombine radiatively** to emit photon.

LED – optical emission

- From the Planck's quantum hypothesis, the wavelength can be related to the energy of photon by

$$\begin{aligned} E &= h\nu, \\ &= \frac{hc}{\lambda}, \\ E \text{ (in eV)} &= \frac{1.24}{\lambda \text{ (in } \mu\text{m)}} \end{aligned} \quad (\text{A})$$

Problem

Prove Equation. A.

Problem

Find the energy of red photon with wavelength 700 nm.

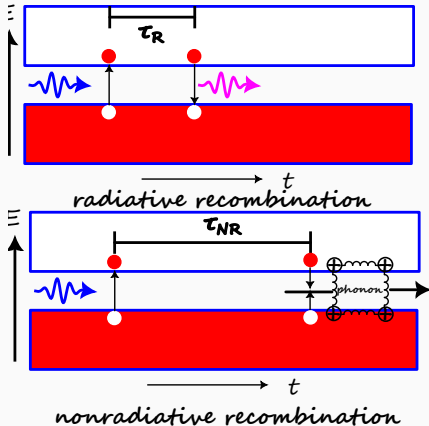
Problem

Find the energy of blue photon with wavelength 400 nm.

Recombination lifetime

Definition

Recombination lifetime is the time period between generation and recombination of charge carriers.



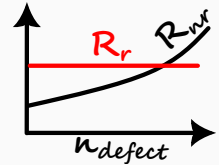
- Consider an electron-hole pair is generated.
- When an electron is in conduction band, its energy has increased. It wants to recombine with a hole to reduce energy.
- The time taken to recombine radiatively τ_r is less than the time taken to recombine non-radiatively τ_{nr} since the defect level lowered the energy of electron.

Recombination rate – Classification of semiconductors

Definition

Recombination rate is defined as the rate at which the excess charge carriers are recombining

- The recombination rate is inversely related to the recombination lifetime.
- In the absence of defects, radiative recombination rate R_r is higher than non-radiative recombination rate R_{nr} .
- However, as the defect density n_{defect} increases, the radiative recombination rate is not affected, but the non-radiative recombination rate increases.
- The bandstructure of the semiconductor also affects the recombination rates. Based on bandstructure the semiconductors classified as –
 - Direct band gap
 - Indirect band gap



LED – efficiency

- The efficiency of LED is defined as ratio of radiative recombination rate R_r to the total recombination rate

$$\eta = \frac{R_r}{R} = \frac{R_r}{R_r + R_{nr}}$$

- Recombination of excess carriers is of two types
 - radiative
 - non-radiative
- Thus, the efficiency is increased when radiative recombination rate increases. However, this is only the internal quantum efficiency.
- External quantum efficiency determines how many generated photons actually are emitted from the semiconductor. External quantum efficiency is degraded by
 - Photon reabsorption
 - Fresnel loss
 - Critical angle loss

- Night lighting
- Traffic signals
- Optical fibre communications
- Medical imaging

Summary of opto-electronic devices

- From the perspective of bias

Reverse bias	Zero bias	Forward bias
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Photo-diode	Solar cell	LED
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- From the perspective of depletion region width

Larger width	Nominal width	Smaller width
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Low current	Nominal current	Large current
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- From the perspective of power consumption

Low Power	Nominal Power	High Power
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Consumption	Generation	Consumption
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- From the perspective of capacitance

Low Capacitance	Nominal Capacitance	High Capacitance
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High frequency	—	—
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End of M2U2